Ben Carnes

Software Developer — Statistics Major, University of Waterloo

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GitHub — LinkedIn **Portfolio**

Technical Skills

Languages: Python, C++, JavaScript, HTML/CSS, SAS, PHP
Data: PyTorch, LangChain, Azure OpenAI, Qdrant
Tools: React, PySide6, Git, Bash, Linux, Docker, Jupyter
Cloud: Azure, CI/CD, OWASP ZAP

Experience

Software Developer Intern Jan 2026 – April 2026
Statistics Canada *Ottawa, ON*

- Implemented 4 reusable modules for generating derived variables for an internal data transformation platform.
- Developed and debugged custom Blockly blocks for visual data transformation workflows.
- Created a working automated OWASP ZAP security scanning reference DevOps pipeline, providing teams with a reusable example for CI/CD vulnerability checks.
- Refactored 3 legacy SAS workflows to Python, improving maintainability and long term extensibility.
- Fixed bugs across internal tools and helped teammates debug implementation issues.

Full Stack Software Developer Intern Apr 2025 – Aug 2025
LC Institute *Waterloo, ON*

- Designed an end-to-end RAG chatbot using Azure OpenAI and Qdrant, giving the nonprofit a natural language Q&A system from scratch.
- Built the chatbot backend orchestrator for handling user queries, then deployed it with Docker and Azure Container Apps and added content safety checks.
- Automated monthly vector ingestion with Azure Container Instances and documented the system architecture.
- Kept projected cloud costs under $200/month while planning for up to 500 concurrent users.
- Deployed and secured the chatbot’s vector database to support reliable backend search.
- Integrated Microsoft Entra ID authentication and secure query routing into the backend orchestrator for the user-facing WordPress UI, connecting users to the chatbot.

Projects

Gamegrab | *PySide6, EasyOCR, PyTorch, C++/DLL, DeepL API*

- Built a desktop OCR + translation tool for PC games, with a capture-to-translation workflow for Japanese dialogue, menus, and on-screen UI.
- Added cross-resolution capture support and a simple 4-button interface for capturing, refreshing, reading, and translating text.

Autonomous Driving Simulation | *C++, ROS2*

- Implemented 4 ROS2 nodes for a robot sim that used local costmaps, global map memory, and A* path planning to navigate around obstacles.
- Wired up the planner, map memory, and control nodes through ROS2 topics and frame IDs so the robot could replan and move toward a goal.

WGCapture | *C++, DLL, Direct3D, Python, PyPI*

- Published a lightweight Python screen-capture library on PyPI, backed by a custom C++/DLL implementation.
- Used Direct3D frame-pool capture to support fullscreen, borderless, per-window, and high-DPI capture scenarios.

Clubs & Activities

WATonomous | *Autonomous Vehicles Student Design Team* May 2026 – Present

Education

University of Waterloo 2028
Bachelor of Mathematics *Waterloo, ON*